Example 12.15

The probability that a girl, preparing for competitive examination will get a State Government service is 0.12, the probability that she will get a Central Government job is 0.25, and the probability that she will get both is 0.07. Find the probability that (*i*) she will get atleast one of the two jobs (*ii*) she will get only one of the two jobs.

Solution

Let I be the event of getting State Government service and C be the event of getting Central Government job.

Given that $P(I) = 0.12$, $P(C) = 0.25$, and $P(I \cap C) = 0.07$

(*i*) P (at least one of the two jobs) $= P(I \text{ or } C) = P(I \cup C)$

$= P(I) + P(C) - P(I \cap C)$

$= 0.12 + 0.25 - 0.07 = 0.30$

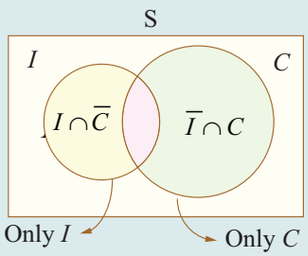
(*ii*) P(only one of the two jobs) $= P$[only I or only C].

$= P(I \cap \overline{C}) + P(\overline{I} \cap C)$

$= \{P(I) - P(I \cap C)\} + \{P(C) - P(I \cap C)\}$

$= \{0.12 - 0.07\} + \{0.25 - 0.07\}$

$= 0.23.$



EXERCISE 12.2

(1) If A and B are mutually exclusive events $P(A) = \frac{3}{8}$ and $P(B) = \frac{1}{8}$, then

find (i) $P(\overline{A})$ (ii) $P(A \cup B)$ (iii) $P(\overline{A} \cap B)$ (iv) $P(\overline{A} \cup \overline{B})$

(2) If A and B are two events associated with a random experiment for which

$P(A) = 0.35$, $P(A \text{ or } B) = 0.85$, and $P(A \text{ and } B) = 0.15$.

Find (i) $P(\text{only } B)$ (ii) $P(\overline{B})$ (iii) $P(\text{only } A)$

(3) A die is thrown twice. Let A be the event, 'First die shows 5' and B be the event, 'second die shows 5'. Find $P(A \cup B)$.

(4) The probability of an event A occurring is 0.5 and B occurring is 0.3. If A and B are mutually exclusive events, then find the probability of

(i) $P(A \cup B)$ (ii) $P\left(A \cap \overline{B}\right)$ (iii) $P\left(\overline{A} \cap B\right)$.

(5) A town has 2 fire engines operating independently. The probability that a fire engine is available when needed is 0.96.

(i) What is the probability that a fire engine is available when needed?

(ii) What is the probability that neither is available when needed?

(6) The probability that a new railway bridge will get an award for its design is 0.48, the probability that it will get an award for the efficient use of materials is 0.36, and that it will get both awards is 0.2. What is the probability, that (i) it will get at least one of the two awards (ii) it will get only one of the awards.

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